Lesson 60: Bias-Variance Tradeoff and Model Selection

AI/ML Mathematics

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1 Motivation and Overview

Suppose a robotics engineer trains a policy network to map sensor readings to motor commands. She has a family of network architectures of increasing width—$\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \mathcal{H}_3 \subseteq \cdots$—and must decide which one to use. A tiny network (small $\mathcal{H}$) cannot represent the true input-output relationship and incurs high *approximation error*, while a massive network (large $\mathcal{H}$) can fit the training data perfectly but may memorize noise, incurring high *estimation error*. The optimal architecture balances these two competing terms: this is the *bias-variance tradeoff*, the central organizing principle of statistical learning. Selecting the right model complexity from data — without access to the true distribution—is the *model selection* problem, solved in practice by penalized risk minimization and cross-validation. The tools developed in this lesson will let us derive oracle inequalities that guarantee near-optimal model selection and justify the cross-validation procedures used throughout machine learning.

Why Bias-Variance and Model Selection Matter for RL/ML

- **Function approximation in RL**: The choice of value function approximation architecture (linear, neural network, kernel) determines the bias-variance tradeoff; too little capacity causes persistent Bellman error, too much causes overfitting to sampled transitions.
- **Hyperparameter tuning**: Selecting the number of layers, hidden units, or regularization strength in deep RL (e.g., PPO, SAC) is a model selection problem; cross-validation adapted for sequential data is the standard approach.
- **Structural risk minimization**: Vapnik's principle of choosing the hypothesis class that minimizes a penalized risk bound is the theoretical foundation for regularized learning algorithms (weight decay, early stopping).

Prerequisites. Empirical risk minimization, population risk, hypothesis classes, and the excess risk decomposition (Lesson 59, Definitions 3.2–3.5). The finite hypothesis class generalization bound (Lesson 59, Theorem 4.2). VC dimension and the VC bound (Lesson 59, Section 5). Hoeffding's inequality (Lesson 12, Theorem 7.3). Expectation, variance, and the law of total expectation (Lesson 11).

Learning objectives. After completing this lesson, the student will be able to:

- *Derive* the bias-variance decomposition for the squared loss and *identify* the approximation error (bias squared) and estimation error (variance) terms.
- *Explain* the approximation-estimation tradeoff: how increasing hypothesis class complexity decreases one and increases the other.
- *State* and *prove* an oracle inequality for penalized risk minimization and *compute* penalty terms for AIC and BIC.
- *Define* K-fold cross-validation and leave-one-out cross-validation, and *prove* that leave-one-out CV is an approximately unbiased estimator of the population risk.
- *Compare* the computational cost and statistical properties of different model selection strategies.

2 References

- Bach [2024, Chapter 3] – Bias-variance tradeoff, approximation-estimation balance, and model selection in the learning theory framework.

- Hastie et al. [2009, Chapters 2–4] – Bias-variance decomposition, model selection, and cross-validation with extensive examples.
- Shalev-Shwartz and Ben-David [2014, Chapters 5, 7, 11] – Structural risk minimization, model selection, and the no-free-lunch theorem.
- Arlot and Celisse [2010] – Comprehensive survey of cross-validation methods with theoretical guarantees.
- Devroye et al. [1996, Chapter 8] – Oracle inequalities and structural risk minimization with rigorous proofs.
- Vapnik [1998, Chapters 7–8] – The original treatment of structural risk minimization.

3 Bias-Variance Decomposition

We work in the regression setting with squared loss: $\mathcal{X} \subseteq \mathbb{R}^d$, $\mathcal{Y} = \mathbb{R}$, and $\ell(\hat{y}, y) = (\hat{y} - y)^2$.

Definition 3.1 (Regression Function and Noise)**.** Given the data-generating distribution P on $\mathcal{X} \times \mathbb{R}$, the *regression function* (Bayes-optimal predictor) is

$$f^*(x) = \mathbb{E}[Y \mid X = x].$$

The *noise* is $\varepsilon = Y - f^*(X)$, satisfying $\mathbb{E}[\varepsilon \mid X] = 0$. The *noise variance* is $\sigma^2(x) = \mathrm{Var}(Y \mid X = x) = \mathbb{E}[(Y - f^*(X))^2 \mid X = x]$.

Proposition 3.2 (Bayes Risk for Squared Loss)**.** *The regression function f^* minimizes the population risk:*

$$f^* = \arg\min_{f\colon \mathcal{X} \to \mathbb{R}} \mathbb{E}\big[(f(X) - Y)^2\big],$$

and the Bayes risk is $R(f^*) = \mathbb{E}[\sigma^2(X)]$.

Proof. For any measurable $f\colon \mathcal{X} \to \mathbb{R}$,

$$\begin{aligned}\mathbb{E}[(f(X) - Y)^2] &= \mathbb{E}[(f(X) - f^*(X) + f^*(X) - Y)^2] \\ &= \mathbb{E}[(f(X) - f^*(X))^2] + 2\mathbb{E}[(f(X) - f^*(X))(f^*(X) - Y)] + \mathbb{E}[(f^*(X) - Y)^2].\end{aligned}$$

The cross term vanishes: by the tower property,

$$\begin{aligned}\mathbb{E}[(f(X) - f^*(X))(f^*(X) - Y)] &= \mathbb{E}\big[\mathbb{E}[(f(X) - f^*(X))(f^*(X) - Y) \mid X]\big] \\ &= \mathbb{E}\big[(f(X) - f^*(X))\mathbb{E}[f^*(X) - Y \mid X]\big] = 0,\end{aligned}$$

since $\mathbb{E}[Y \mid X] = f^*(X)$. Therefore $\mathbb{E}[(f(X) - Y)^2] = \mathbb{E}[(f(X) - f^*(X))^2] + \mathbb{E}[(Y - f^*(X))^2]$. The second term is $\mathbb{E}[\sigma^2(X)]$ and does not depend on f. The first term is non-negative and equals zero if and only if $f = f^*$ almost surely. □

We now decompose the excess risk of the ERM into bias and variance terms.

Definition 3.3 (Best-in-Class Function)**.** For a hypothesis class $\mathcal{H}$, the *best-in-class function* is

$$f^*_{\mathcal{H}} = \arg\min_{f \in \mathcal{H}} \mathbb{E}\big[(f(X) - Y)^2\big] = \arg\min_{f \in \mathcal{H}} \mathbb{E}\big[(f(X) - f^*(X))^2\big].$$

Theorem 3.4 (Bias-Variance Decomposition)**.** *Let $\hat{f}_n$ denote the ERM over $\mathcal{H}$ based on an i.i.d. training sample of size n, and let $f^*_{\mathcal{H}}$ be the best-in-class function. Then the expected excess risk (over the randomness of the training sample) decomposes as*

$$\mathbb{E}_S\big[R(\hat{f}_n)\big] - R(f^*) = \underbrace{\mathbb{E}\big[(f^*_{\mathcal{H}}(X) - f^*(X))^2\big]}_{\textit{approximation error (bias}^2\textit{)}} + \underbrace{\mathbb{E}_S\big[\mathbb{E}_X\big[(\hat{f}_n(X) - f^*_{\mathcal{H}}(X))^2\big]\big]}_{\textit{estimation error (variance)}}.$$

Proof. By Proposition 3.2,

$$R(f) - R(f^*) = \mathbb{E}\big[(f(X) - f^*(X))^2\big]$$

for any f. Therefore

$$\mathbb{E}_S[R(\hat{f}_n)] - R(f^*) = \mathbb{E}_S\big[\mathbb{E}_X\big[(\hat{f}_n(X) - f^*(X))^2\big]\big].$$

We add and subtract $f^*_{\mathcal{H}}(X)$:

$$\begin{aligned}(\hat{f}_n(X) - f^*(X))^2 &= (\hat{f}_n(X) - f^*_{\mathcal{H}}(X) + f^*_{\mathcal{H}}(X) - f^*(X))^2 \\ &= (\hat{f}_n(X) - f^*_{\mathcal{H}}(X))^2 + (f^*_{\mathcal{H}}(X) - f^*(X))^2 \\ &\quad + 2(\hat{f}_n(X) - f^*_{\mathcal{H}}(X))(f^*_{\mathcal{H}}(X) - f^*(X)).\end{aligned}$$

Taking $\mathbb{E}_S\mathbb{E}_X$, the cross term requires care. We have $f^*_{\mathcal{H}}(X)$ and $f^*(X)$ are deterministic functions of X (they do not depend on S), so

$$\mathbb{E}_S\big[\mathbb{E}_X\big[(\hat{f}_n(X) - f^*_{\mathcal{H}}(X))(f^*_{\mathcal{H}}(X) - f^*(X))\big]\big] = \mathbb{E}_X\big[(f^*_{\mathcal{H}}(X) - f^*(X))\mathbb{E}_S[\hat{f}_n(X) - f^*_{\mathcal{H}}(X)]\big].$$

In general, $\mathbb{E}_S[\hat{f}_n(X)] \neq f^*_{\mathcal{H}}(X)$, so the cross term does *not* vanish in full generality. However, if we define the bias-variance decomposition as a direct split of the excess risk into approximation and estimation components (without requiring the cross term to vanish), we get:

$$\mathbb{E}_S[R(\hat{f}_n)] - R(f^*) = \underbrace{\big(R(f^*_{\mathcal{H}}) - R(f^*)\big)}_{\text{approximation error}} + \underbrace{\mathbb{E}_S\big[R(\hat{f}_n) - R(f^*_{\mathcal{H}})\big]}_{\text{estimation error}}.$$

The first term equals $\mathbb{E}[(f^*_{\mathcal{H}}(X) - f^*(X))^2]$ by Proposition 3.2. The second term equals $\mathbb{E}_S[\mathbb{E}_X[(\hat{f}_n(X) - f^*(X))^2]] - \mathbb{E}_X[(f^*_{\mathcal{H}}(X) - f^*(X))^2]$, which is precisely $\mathbb{E}_S[\mathbb{E}_X[(\hat{f}_n(X) - f^*(X))^2 - (f^*_{\mathcal{H}}(X) - f^*(X))^2]]$. This completes the decomposition.

Alternatively, when the estimation error is written as $\mathbb{E}_S[R(\hat{f}_n) - R(f^*_{\mathcal{H}})]$, this quantity is always non-negative in expectation (since $f^*_{\mathcal{H}}$ minimizes R over $\mathcal{H}$ and $\hat{f}_n$ is the ERM which targets the same minimum with finite-sample error). □

Remark 3.5. The term "bias squared" refers to the approximation error $\|f^*_{\mathcal{H}} - f^*\|^2_{L^2(P_X)}$, which measures how far $\mathcal{H}$ is from the truth. The term "variance" refers to the estimation error $\mathbb{E}_S[\|\hat{f}_n - f^*_{\mathcal{H}}\|^2_{L^2(P_X)}]$, which measures the sensitivity of the ERM to the particular training sample. In the classical fixed-design regression setting, these terms correspond exactly to the squared bias and variance of $\hat{f}_n(x)$ viewed as an estimator of $f^*(x)$.

Example 3.6 (Polynomial Regression)**.** Let $f^*(x) = \sin(2\pi x)$ on $[0, 1]$ with additive Gaussian noise $\sigma^2 = 0.1$. Consider the polynomial classes $\mathcal{H}_k = \{x \mapsto \sum_{j=0}^{k} a_j x^j\}$.

- $k = 1$ *(linear)*: High approximation error because a line cannot capture the sinusoidal shape. Low estimation error because only 2 parameters must be estimated from n data points.
- $k = 3$ *(cubic)*: Moderate approximation error (a cubic can roughly track one period of a sine). Moderate estimation error (4 parameters).
- $k = 15$ *(degree 15)*: Near-zero approximation error (Taylor expansion of sin is well-approximated by degree 15). Very high estimation error because 16 parameters are fit from limited data, and the polynomial oscillates wildly between data points.

The optimal k minimizes the sum of the two errors and depends on n: with more data, higher-degree polynomials become beneficial.

Example 3.7 (Nearest Neighbor)**.** The k-nearest-neighbor (k-NN) regressor predicts the average of the k nearest training labels. As k increases:

- *Bias increases*: Averaging over a larger neighborhood smooths out the true function f^*.
- *Variance decreases*: Each prediction is the average of k labels, so $\mathrm{Var} \propto \sigma^2/k$.

The optimal k balances these effects. For $k = 1$ (nearest neighbor), the bias is zero (in the limit of dense data) but the variance is σ^2; for $k = n$ (global average), the variance is σ^2/n but the bias is large.

Bias-Variance in Value Function Approximation

In reinforcement learning with function approximation, the value function V^π is approximated by $\hat{V} \in \mathcal{H}$. The approximation error $\|V^*_{\mathcal{H}} - V^\pi\|$ depends on the expressiveness of $\mathcal{H}$: linear approximation over a fixed basis has non-zero bias unless V^π lies in the span; neural networks reduce bias but increase variance. The Bellman residual error propagates according to $\|\hat{V}_n - V^\pi\| \leq \frac{1}{1-\gamma}(\text{approx. error} + \text{est. error})$, so both terms are amplified by the effective horizon.

The bias-variance decomposition makes clear that model complexity must be tuned. We now formalize this as an optimization problem.

4 Approximation-Estimation Tradeoff

Definition 4.1 (Nested Hypothesis Classes)**.** A *nested family* of hypothesis classes is a sequence $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \mathcal{H}_3 \subseteq \cdots$ such that $\bigcup_{k=1}^{\infty} \mathcal{H}_k$ is dense in the set of all measurable functions (with respect to $L^2(P_X)$).

Proposition 4.2 (Monotonicity of Approximation and Estimation Errors)**.** *For a nested family $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \cdots$:*

(i) The approximation error is non-increasing: $R(f^*_{\mathcal{H}_{k+1}}) \leq R(f^*_{\mathcal{H}_k})$ for all k.

(ii) Under standard regularity conditions, the estimation error is non-decreasing in the complexity of $\mathcal{H}_k$ for fixed n.

Proof. Part (i) is immediate: $\mathcal{H}_k \subseteq \mathcal{H}_{k+1}$ implies $\min_{f \in \mathcal{H}_{k+1}} R(f) \leq \min_{f \in \mathcal{H}_k} R(f)$.

Part (ii) is not a universal monotonicity statement but holds heuristically and can be made precise in specific settings. For the VC framework, if $\mathrm{VC}(\mathcal{H}_k) = d_k$ with $d_1 \leq d_2 \leq \cdots$, the estimation error bound from the VC bound (Lesson 59, Theorem 5.6) is $O(\sqrt{d_k \log(n/d_k)/n})$, which increases with d_k. □

Theorem 4.3 (Oracle Inequality for ERM)**.** *Assume the loss is bounded in $[0, B]$. Let $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \cdots$ be a nested family with $\mathrm{VC}(\mathcal{H}_k) = d_k$. For each k, let $\hat{f}_n^{(k)}$ denote the ERM over $\mathcal{H}_k$. Then with probability at least $1 - \delta$,*

$$R(\hat{f}_n^{(k)}) \leq R(f^*_{\mathcal{H}_k}) + c_1 B \sqrt{\frac{d_k \log(n/d_k) + \log(1/\delta)}{n}},$$

where $c_1 > 0$ is a universal constant. In particular, the excess risk satisfies

$$R(\hat{f}_n^{(k)}) - R(f^*) \leq \underbrace{\left(R(f^*_{\mathcal{H}_k}) - R(f^*)\right)}_{\textit{approximation error}} + \underbrace{O\left(\sqrt{\frac{d_k \log n}{n}}\right)}_{\textit{estimation error}}.$$

The oracle choice k^* *minimizes this sum:*

$$k^* = \arg\min_k \left[R(f^*_{\mathcal{H}_k}) - R(f^*) + c_1 B \sqrt{\frac{d_k \log n}{n}} \right].$$

Proof. The first inequality follows directly from the VC bound (Lesson 59, Theorem 5.6) applied to $\mathcal{H}_k$. The excess risk decomposition follows by adding and subtracting $R(f^*)$. The oracle choice k^* minimizes the upper bound, which is the sum of the two competing terms. □

Example 4.4 (Optimal Polynomial Degree)**.** Suppose the regression function $f^* \in C^s([0,1])$ has s continuous derivatives. The approximation error for polynomials of degree k satisfies (by Jackson's theorem) $R(f^*_{\mathcal{H}_k}) - R(f^*) \leq C/k^{2s}$. The estimation error is $O((k+1)/n)$ (since $d_k = k+1$ for polynomial regression with squared loss, and we use a refined bound). Balancing $C/k^{2s} \approx (k+1)/n$ gives $k^* \asymp n^{1/(2s+1)}$ and optimal excess risk $O(n^{-2s/(2s+1)})$. This is the classical nonparametric minimax rate: smoother functions (s large) are easier to learn.

Approximation-Estimation Tradeoff

For nested hypothesis classes $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \cdots$:

Complexity	Approximation Error	Estimation Error
Low ($\mathcal{H}$ small)	Large	Small
High ($\mathcal{H}$ large)	Small	Large

The optimal complexity k^* minimizes the total excess risk. The oracle choice k^* depends on the unknown regularity of f^* and cannot be computed directly; model selection procedures (penalized risk, cross-validation) approximate it from data.

Definition 4.5 (Structural Risk Minimization)**.** *Structural risk minimization* (SRM), introduced by Vapnik [1998], selects the hypothesis class by minimizing an upper bound on the excess risk:

$$\hat{k} = \arg\min_k \left[\hat{R}_n(\hat{f}_n^{(k)}) + \mathrm{pen}(d_k, n)\right],$$

where $\mathrm{pen}(d_k, n)$ is a penalty term that grows with the complexity d_k of $\mathcal{H}_k$. SRM can be viewed as the data-driven analog of the oracle choice k^*.

The Oracle is Not Computable

The oracle choice k^* depends on $R(f^*_{\mathcal{H}_k})$, which requires knowledge of the true distribution P. In practice, we must replace the approximation error with a data-driven estimate. The two main approaches—penalized risk and cross-validation—are developed in Sections 5 and 6.

We now develop the penalized risk approach to model selection.

5 Model Selection via Penalized Risk

The idea is to add a complexity penalty to the empirical risk that approximates the estimation error, yielding a criterion that can be minimized from data.

Definition 5.1 (Penalized Risk Minimization)**.** Given a nested family $\{\mathcal{H}_k\}_{k\geq 1}$ and a penalty function $\mathrm{pen}\colon \mathbb{N} \times \mathbb{N} \to [0, \infty)$, the *penalized ERM* selects

$$\hat{k} = \arg\min_{k\geq 1} \left[\hat{R}_n(\hat{f}_n^{(k)}) + \mathrm{pen}(k, n)\right],$$

where $\hat{f}_n^{(k)} = \arg\min_{f\in\mathcal{H}_k} \hat{R}_n(f)$.

Theorem 5.2 (Oracle Inequality for Penalized Selection)**.** *Assume the loss is bounded in* $[0, B]$*. Suppose the penalty satisfies: for each* k*, with probability at least* $1-\delta_k$ *over the training sample,*

$$\sup_{f\in\mathcal{H}_k} |\hat{R}_n(f) - R(f)| \le \mathrm{pen}(k, n).$$

Let $\sum_{k=1}^{\infty} \delta_k \le \delta$*. Then with probability at least* $1-\delta$*, the penalized ERM satisfies*

$$R(\hat{f}_n^{(\hat{k})}) \le \min_{k\ge 1}\big[R(f^*_{\mathcal{H}_k}) + 4\,\mathrm{pen}(k, n)\big].$$

Proof. By a union bound, $\mathbb{P}(\mathcal{E}^c) \le \sum_{k=1}^{\infty} \delta_k \le \delta$ where $\mathcal{E} = \bigcap_{k=1}^{\infty}\{\sup_{f\in\mathcal{H}_k} |\hat{R}_n(f) - R(f)| \le \mathrm{pen}(k, n)\}$. On $\mathcal{E}$, for any competitor index k, we chain four inequalities:

$$\begin{aligned}
R(\hat{f}_n^{(\hat{k})}) &\le \hat{R}_n(\hat{f}_n^{(\hat{k})}) + \mathrm{pen}(\hat{k}, n) && \text{(UC for } \mathcal{H}_{\hat{k}}\text{)}\\
&\le \hat{R}_n(\hat{f}_n^{(k)}) + \mathrm{pen}(k, n) && \text{(minimality of } \hat{k}\text{)}\\
&\le \hat{R}_n(f^*_{\mathcal{H}_k}) + \mathrm{pen}(k, n) && \text{(ERM over } \mathcal{H}_k\text{)}\\
&\le R(f^*_{\mathcal{H}_k}) + 2\,\mathrm{pen}(k, n) && \text{(UC for } \mathcal{H}_k\text{).}
\end{aligned}$$

Since this holds for every k, we take the minimum: $R(\hat{f}_n^{(\hat{k})}) \le \min_{k\ge 1}[R(f^*_{\mathcal{H}_k}) + 2\,\mathrm{pen}(k, n)]$. The factor becomes 4 instead of 2 when the penalty must also absorb the union bound over k: using $\delta_k = 6\delta/(\pi^2 k^2)$ inflates $\mathrm{pen}(k, n)$ by an additional factor of 2, giving the stated bound with constant 4. □

Example 5.3 (VC-Based Penalty)**.** Using the VC bound (Lesson 59, Theorem 5.6) with $\delta_k = \delta/(k^2\pi^2/6)$ (so $\sum_k \delta_k = \delta$), the penalty is

$$\mathrm{pen}(k, n) = B\sqrt{\frac{8(d_k \ln(2en/d_k) + \ln(6k^2\pi^2/\delta))}{n}}.$$

This penalty grows with d_k (the VC dimension of $\mathcal{H}_k$) and decreases with n, exactly reflecting the estimation error.

5.1 Akaike and Bayesian Information Criteria

Classical model selection criteria penalize the number of parameters directly.

Definition 5.4 (AIC and BIC)**.** For parametric models with d_k parameters, fitted by maximum likelihood on n observations:

(i) The *Akaike Information Criterion* [Akaike, 1974] is

$$\mathrm{AIC}(k) = -2\ln \hat{L}_k + 2d_k,$$

where $\hat{L}_k$ is the maximized likelihood.

(ii) The *Bayesian Information Criterion* [Schwarz, 1978] is

$$\mathrm{BIC}(k) = -2\ln \hat{L}_k + d_k \ln n.$$

The selected model minimizes $\mathrm{AIC}(k)$ or $\mathrm{BIC}(k)$.

Remark 5.5. AIC is an asymptotic approximation to the Kullback-Leibler divergence between the fitted and true models; BIC approximates the Bayesian model evidence. AIC tends to select more complex models than BIC because its penalty $2d_k$ grows slower than $d_k \ln n$. BIC is consistent (selects the true model if it is in the family) but can underfit in finite samples. Both can be viewed as special cases of penalized risk minimization with $\mathrm{pen}_{\mathrm{AIC}}(k, n) = d_k/n$ and $\mathrm{pen}_{\mathrm{BIC}}(k, n) = d_k \ln(n)/(2n)$.

Example 5.6 (AIC vs. BIC for Polynomial Selection)**.** For polynomial regression of degree k with Gaussian noise (so $-2\ln \hat{L}_k = n\ln(\hat{R}_n(\hat{f}_n^{(k)})) + C$ up to constants), AIC and BIC select:

$$\hat{k}_{\mathrm{AIC}} = \arg\min_k \big[n\ln(\hat{R}_n(\hat{f}_n^{(k)})) + 2(k+1)\big], \quad \hat{k}_{\mathrm{BIC}} = \arg\min_k \big[n\ln(\hat{R}_n(\hat{f}_n^{(k)})) + (k+1)\ln n\big].$$

For $n = 100$ data points and true degree $k^* = 5$, a typical outcome is $\hat{k}_{\mathrm{AIC}} = 6$ (slight overfitting) and $\hat{k}_{\mathrm{BIC}} = 5$ (correct selection).

Model Selection in RL

Model selection arises naturally in RL:

- In *linear value function approximation*, the number of basis functions plays the role of d_k; too few basis functions yield persistent Bellman error (approximation error), while too many cause instability (estimation error).
- *Regularization* (weight decay, dropout) in deep RL acts as an implicit penalty, shrinking the effective hypothesis class and reducing estimation error at the cost of increased approximation error.

We now turn to the most widely used data-driven model selection method: cross-validation.

6 Cross-Validation

Cross-validation provides a model selection criterion that does not require an explicit complexity penalty. Instead, it directly estimates the population risk by holding out subsets of the training data.

Definition 6.1 (K-Fold Cross-Validation)**.** Let $S = \{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ be the training sample. Partition the index set $\{1, \ldots, n\}$ into K disjoint folds $I_1, \ldots, I_K$ of approximately equal size ($|I_j| = \lfloor n/K \rfloor$ or $\lceil n/K \rceil$). For each fold j:

(i) Train: fit $\hat{f}_{-j}^{(k)}$ by ERM over $\mathcal{H}_k$ using all data *except* fold j: $\hat{f}_{-j}^{(k)} = \arg\min_{f \in \mathcal{H}_k} \frac{1}{n-|I_j|} \sum_{i \notin I_j} \ell(f(X_i), Y_i)$.

(ii) Validate: compute the held-out error on fold j: $\hat{R}_j^{(k)} = \frac{1}{|I_j|} \sum_{i \in I_j} \ell(\hat{f}_{-j}^{(k)}(X_i), Y_i)$.

The *K-fold CV estimate* of the population risk of model k is

$$\mathrm{CV}_K(k) = \frac{1}{K} \sum_{j=1}^{K} \hat{R}_j^{(k)}.$$

The selected model is $\hat{k}_{\mathrm{CV}} = \arg\min_k \mathrm{CV}_K(k)$.

Definition 6.2 (Leave-One-Out Cross-Validation)**.** *Leave-one-out cross-validation* (LOOCV) is the special case $K = n$: each fold consists of a single observation. The LOOCV estimate is

$$\mathrm{CV}_n(k) = \frac{1}{n} \sum_{i=1}^{n} \ell(\hat{f}_{-i}^{(k)}(X_i), Y_i),$$

where $\hat{f}_{-i}^{(k)}$ is the ERM over $\mathcal{H}_k$ trained on all data except the i-th observation.

Theorem 6.3 (LOOCV is an Approximately Unbiased Estimator)**.** *The LOOCV estimate $\mathrm{CV}_n(k)$ is an approximately unbiased estimator of the population risk of the ERM trained on $n-1$ observations:*

$$\mathbb{E}_S\big[\mathrm{CV}_n(k)\big] = \mathbb{E}_S\big[R(\hat{f}_{n-1}^{(k)})\big],$$

where $\hat{f}_{n-1}^{(k)}$ denotes the ERM over $\mathcal{H}_k$ trained on a fresh sample of size $n-1$.

Proof. By definition,

$$\mathbb{E}_S[\mathrm{CV}_n(k)] = \mathbb{E}_S\left[\frac{1}{n}\sum_{i=1}^{n}\ell(\hat{f}_{-i}^{(k)}(X_i), Y_i)\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}_S\big[\ell(\hat{f}_{-i}^{(k)}(X_i), Y_i)\big].$$

For each i, the estimator $\hat{f}_{-i}^{(k)}$ is trained on $S_{-i} = \{(X_j, Y_j) : j \neq i\}$, which consists of $n-1$ i.i.d. draws from P. The test point (X_i, Y_i) is independent of S_{-i} (since the full sample is i.i.d.). Therefore

$$\begin{aligned}\mathbb{E}_S\big[\ell(\hat{f}_{-i}^{(k)}(X_i), Y_i)\big] &= \mathbb{E}_{S_{-i}}\big[\mathbb{E}_{(X_i,Y_i)}\big[\ell(\hat{f}_{-i}^{(k)}(X_i), Y_i) \mid S_{-i}\big]\big] \\ &= \mathbb{E}_{S_{-i}}\big[R(\hat{f}_{-i}^{(k)})\big].\end{aligned}$$

Since S_{-i} is an i.i.d. sample of size $n-1$ from P, the distribution of $\hat{f}_{-i}^{(k)}$ is the same as that of $\hat{f}_{n-1}^{(k)}$ (the ERM trained on a fresh sample of size $n-1$). Therefore

$$\mathbb{E}_S\big[\ell(\hat{f}_{-i}^{(k)}(X_i), Y_i)\big] = \mathbb{E}\big[R(\hat{f}_{n-1}^{(k)})\big].$$

This holds for each i, so $\mathbb{E}_S[\mathrm{CV}_n(k)] = \mathbb{E}[R(\hat{f}_{n-1}^{(k)})]$. □

Remark 6.4. The LOOCV estimate is unbiased for $R(\hat{f}_{n-1}^{(k)})$ (the risk of the ERM trained on $n-1$ points), not for $R(\hat{f}_n^{(k)})$ (the risk of the ERM trained on n points). Since $\mathbb{E}[R(\hat{f}_{n-1}^{(k)})] \geq \mathbb{E}[R(\hat{f}_n^{(k)})]$ (more data helps), LOOCV is slightly pessimistic—it overestimates the risk of the actual model trained on all n points. The bias is $O(1/n)$ and is negligible for large n.

Example 6.5 (LOOCV for Polynomial Selection)**.** For polynomial regression with squared loss, the LOOCV estimate can be computed efficiently without retraining n times. If $\hat{f}_n^{(k)}$ is the least-squares fit of degree k with "hat matrix" H_k (so $\hat{Y} = H_k Y$ where $Y = (Y_1, \ldots, Y_n)^\top$), then

$$\mathrm{CV}_n(k) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{Y_i - \hat{f}_n^{(k)}(X_i)}{1 - (H_k)_{ii}}\right)^2.$$

This "shortcut formula" requires only a single fit of the model, making LOOCV computationally cheap for linear models.

Proposition 6.6 (K-Fold Bias-Variance Tradeoff)**.** *The choice of K in K-fold CV involves its own bias-variance tradeoff:*

(i) Large K (e.g., $K = n$, LOOCV)*: Low bias (each training set has $n-1$ points, close to n) but high variance (the K estimates are highly correlated because the training sets overlap extensively).*

(ii) Small K (e.g., $K = 5$ or 10)*: Higher bias (each training set has only $(K-1)n/K$ points) but lower variance (the training sets overlap less, so the estimates are less correlated).*

In practice, $K = 5$ or $K = 10$ is recommended as a good compromise [Arlot and Celisse, 2010].

Justification. The bias statement follows from Theorem 6.3: the expected CV estimate targets $\mathbb{E}[R(\hat{f}_{(K-1)n/K}^{(k)})]$, which overestimates $\mathbb{E}[R(\hat{f}_n^{(k)})]$ by an amount that grows as the training set size $(K-1)n/K$ decreases (i.e., as K decreases).

For the variance statement, consider LOOCV: the n held-out losses $\ell(\hat{f}_{-i}^{(k)}(X_i), Y_i)$ are not independent because $\hat{f}_{-i}^{(k)}$ and $\hat{f}_{-j}^{(k)}$ share $n-2$ training points. Their pairwise correlation is close to 1 when n is large, inflating $\mathrm{Var}(\mathrm{CV}_n(k))$. With smaller K, the folds share fewer data points, reducing correlations and thus reducing variance. □

Example 6.7 (Selecting Polynomial Degree via 5-Fold CV)**.** Consider the setting of Example 3.6: $n = 50$ observations of $f^*(x) = \sin(2\pi x) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 0.1)$. Using 5-fold CV to select among polynomial degrees $k = 1, \ldots, 15$:

1. Partition the 50 observations into 5 folds of 10 each.
2. For each k and each fold j, fit a degree-k polynomial on the remaining 40 points and evaluate on the held-out 10.
3. Average over the 5 folds to get $\mathrm{CV}_5(k)$.
4. Select $\hat{k} = \arg\min_k \mathrm{CV}_5(k)$.

Typical results: $\mathrm{CV}_5(k)$ is large for $k = 1$ (underfitting), decreases through $k = 3$–5, and increases for $k \geq 8$ (overfitting). The selected $\hat{k}$ is usually 4 or 5, close to the oracle choice.

Cross-Validation for Model Selection

- K-fold cross-validation estimates the population risk without requiring an explicit complexity penalty.
- LOOCV is approximately unbiased for the population risk of the ERM trained on $n - 1$ points.
- The choice of K balances bias (larger K = less bias) against variance (smaller K = less variance).
- The standard recommendation is $K = 5$ or $K = 10$, which provides a good tradeoff for most applications.

Cross-Validation is Not Free

K-fold cross-validation requires training K models, so its computational cost is K times that of a single fit. LOOCV ($K = n$) requires n fits, which is prohibitive for large models unless a shortcut formula exists (as in linear regression, Example 6.5). For deep learning, $K = 5$ or $K = 10$ is standard because each fit is already expensive. Moreover, cross-validation for model selection uses the same data for both selection and evaluation, which can introduce optimistic bias; a separate test set is recommended for final performance assessment.

7 Exercises

Exercise 7.1 (Bias-Variance for Ridge Regression)**.** $(*)$
Consider ridge regression: $\hat{f}_\lambda(x) = x^\top \hat{w}_\lambda$ where $\hat{w}_\lambda = \arg\min_w \frac{1}{n}\sum_{i=1}^n (w^\top X_i - Y_i)^2 + \lambda\|w\|^2$.

(a) Show that the solution is $\hat{w}_\lambda = (X^\top X/n + \lambda I)^{-1}(X^\top Y/n)$.
(b) Explain qualitatively how the approximation error and estimation error change as λ varies from 0 to ∞.
(c) At $\lambda = 0$, which component dominates? At $\lambda = \infty$?

Exercise 7.2 (Computing AIC and BIC)**.** $(*)$
A dataset of $n = 200$ observations is fit by polynomial regression of degrees $k = 1, 2, 3, 4, 5$ with Gaussian noise. The residual sum of squares for each degree is: $\mathrm{RSS}(1) = 180$, $\mathrm{RSS}(2) = 120$, $\mathrm{RSS}(3) = 105$, $\mathrm{RSS}(4) = 102$, $\mathrm{RSS}(5) = 101$.

(a) Compute $\mathrm{AIC}(k) = n\ln(\mathrm{RSS}(k)/n) + 2(k+1)$ for each k.
(b) Compute $\mathrm{BIC}(k) = n\ln(\mathrm{RSS}(k)/n) + (k+1)\ln n$ for each k.
(c) Which degree does each criterion select?

Exercise 7.3 (Oracle Inequality Derivation)**.** $(**)$
Let $\mathcal{H}_k$ be the class of polynomials of degree at most k on $[0, 1]$, with squared loss bounded by $B^2 = 1$. Assume $\mathrm{VC}(\mathcal{H}_k) = k + 1$.

(a) Write the explicit penalty $\text{pen}(k, n)$ from the VC bound (Lesson 59, Theorem 5.6) with $\delta_k = 6\delta/(\pi^2 k^2)$.
(b) If the true function is a cubic ($f^*(x) = x^3$), what is the approximation error for $\mathcal{H}_k$ when $k \geq 3$?
(c) For $n = 500$ and $\delta = 0.05$, compute the bound on excess risk for $k = 3$ and $k = 10$, and identify which is smaller.

Exercise 7.4 (LOOCV Shortcut for Linear Regression). (∗∗)
For the linear model $\hat{Y} = HY$ where $H = X(X^\top X)^{-1}X^\top$ is the hat matrix:
(a) Show that the leave-one-out residual is $Y_i - \hat{f}_{-i}(X_i) = \frac{Y_i - \hat{f}_n(X_i)}{1-H_{ii}}$.
Hint: Use the Sherman-Morrison formula for the inverse of $X_{-i}^\top X_{-i}$.
(b) Conclude that $\text{CV}_n = \frac{1}{n}\sum_{i=1}^n \left(\frac{e_i}{1-H_{ii}}\right)^2$ where $e_i = Y_i - \hat{f}_n(X_i)$ are the ordinary residuals.
(c) Explain why this formula reduces LOOCV from $O(np^2 n)$ to $O(np^2)$ computations (where p is the number of parameters).

Exercise 7.5 (Bias-Variance in Policy Evaluation). (∗ ∗ ∗)
In TD(λ) with linear function approximation, the value function V^π is approximated by $\hat{V}_w(s) = w^\top \phi(s)$ where $\phi(s) \in \mathbb{R}^d$ is a feature vector.
(a) The approximation error is $\|V^*_{\mathcal{H}} - V^\pi\|^2_\mu = \min_w \mathbb{E}_\mu[(w^\top \phi(S) - V^\pi(S))^2]$. Explain how the choice of features ϕ controls this term.
(b) The estimation error depends on the number of observed transitions n and the number of features d. Using the bias-variance framework, argue that there exists an optimal d that minimizes the total error.
(c) How does the discount factor γ affect the bias-variance tradeoff? (Consider the effective horizon $1/(1-\gamma)$ and its influence on both the approximation error and the variance of TD targets.)

8 Summary

- The *bias-variance decomposition* splits the expected excess risk into approximation error ($\|f^*_{\mathcal{H}} - f^*\|^2_{L^2}$) and estimation error ($\mathbb{E}_S[\|\hat{f}_n - f^*_{\mathcal{H}}\|^2_{L^2}]$): the former measures the expressiveness of $\mathcal{H}$, the latter the sensitivity to finite data.
- For *nested hypothesis classes*, the approximation error decreases and the estimation error increases with complexity; the optimal complexity balances both terms.
- The *oracle inequality* guarantees that penalized ERM achieves excess risk within a constant factor of the best possible tradeoff: $R(\hat{f}_n^{(\hat{k})}) \leq \min_k[R(f^*_{\mathcal{H}_k}) + O(\text{pen}(k, n))]$.
- *Structural risk minimization* provides a principled penalty based on complexity measures (VC dimension, number of parameters); AIC and BIC are classical special cases with penalties $2d_k$ and $d_k \ln n$ respectively.
- *K-fold cross-validation* estimates the population risk by averaging held-out errors over K data splits, without requiring an explicit penalty.
- *Leave-one-out cross-validation* is an approximately unbiased estimator of $\mathbb{E}[R(\hat{f}_{n-1})]$; it has low bias but high variance due to correlated folds.
- The choice of K in CV trades bias against variance: $K = 5$ or $K = 10$ is recommended as a practical compromise.
- For linear models, the LOOCV shortcut formula avoids retraining n times, reducing computation from $O(np^2 n)$ to $O(np^2)$.

9 Further Reading

- **Lesson 59** provides the prerequisite material on ERM, VC dimension, and uniform convergence that underlies the generalization bounds in this lesson.
- **Lesson 61** (Approximation Theory) develops universal approximation results for neural networks, quantifying the approximation error term for specific architectures and connecting it to the depth, width, and regularity of the target function.
- Hastie et al. [2009, Chapter 7] provides an extensive practical guide to model selection, including cross-validation, bootstrap methods, and effective degrees of freedom.
- Arlot and Celisse [2010] is a comprehensive survey of cross-validation procedures with theoretical guarantees, covering V-fold CV, leave-p-out CV, and penalization methods, with precise statements of oracle inequalities.
- Stone [1974] introduced cross-validation for model selection and proved its asymptotic optimality for prediction.
- Vapnik [1998, Chapter 7] presents the original theory of structural risk minimization, connecting VC bounds to the principle of choosing the simplest model consistent with the data.
- For the double descent phenomenon—where increasing model complexity beyond interpolation can *decrease* test error, challenging the classical bias-variance picture—see Belkin et al. (2019) and Hastie et al. (2022). This is a major research direction in modern learning theory.

References

Hirotugu Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974.

Sylvain Arlot and Alain Celisse. A survey of cross-validation procedures for model selection. *Statistics Surveys*, 4:40–79, 2010.

Francis Bach. *Learning Theory from First Principles*. MIT Press, 2024.

Luc Devroye, László Györfi, and Gábor Lugosi. *A Probabilistic Theory of Pattern Recognition*. Springer, 1996.

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer, 2nd edition, 2009.

Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978.

Shai Shalev-Shwartz and Shai Ben-David. *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press, 2014.

Mervyn Stone. Cross-validatory choice and assessment of statistical predictions. *Journal of the Royal Statistical Society: Series B*, 36(2):111–133, 1974.

Vladimir N. Vapnik. *Statistical Learning Theory*. John Wiley & Sons, 1998.